

Course Project Discription

Overview

The Course Project is an opportunity for you to apply what you have learned in class to a problem of your interest. Potential projects usually fall into these two tracks:

- **Applications.** If you're coming to the class with a specific background and interests (e.g. biology, engineering, physics), we'd love to see you apply various deep learning architectures to problems related to your particular domain of interest. Pick a real-world problem and apply deep learning to solve it.
- **Models.** You can build a new model, or a new variant of existing models, and apply it to tackle various tasks. This track might be more challenging, and sometimes leads to a piece of publishable work.

To inspire ideas, you might also look at recent deep learning publications from top-tier conferences, as well as other resources below.

- CVPR: IEEE Conference on Computer Vision and Pattern Recognition
- ICCV: International Conference on Computer Vision
- ECCV: European Conference on Computer Vision
- NIPS: Neural Information Processing Systems
- ICLR: International Conference on Learning Representations
- ICML: International Conference on Machine Learning
- Awesome Deep Vision
  Kaggle challenges: An online machine learning competition website. For example, a Yelp classification challenge.

For applications, this type of projects would involve careful data preparation, an appropriate loss function, details of training and cross-validation and good test set evaluations and model comparisons. Don't be afraid to think outside of the box. Some successful examples can be found below:

- Teaching Deep Convolutional Neural Networks to Play Go
- Playing Atari with Deep Reinforcement Learning
- Winning the Galaxy Challenge with convnets
- Recommending music on Spotify with deep learning

In the link below, you might also get inspiration by taking a look at public datasets:

- Awesome Public Datasets

Collaboration Policy

You can work in teams of up to **3** people. We do expect that projects done with 3 people have more impressive writeup and results than projects done with 2 people. To get a sense for the scope and expectations for 2–people projects have a look at project reports from previous years.

Honor Code

You may consult any papers, books, online references, or publicly available implementations for ideas and code that you may want to incorporate into your strategy or algorithm, so long as you clearly cite your sources in your code and your writeup. However, under no circumstances may you look at another group's code or incorporate their code into your project.

If you are combining your course project with the project from another class, you must receive permission from the instructors, and clearly explain in the Proposal, Milestone, and Final Report the exact portion of the project that is being counted for this course. In this case you must prepare separate reports for each course, and submit your final report for the other course as well.

Important Dates

Unless otherwise noted, all project items are due by 11:59 pm.

- Project proposal: due Monday, 5 Oct 2026.
- Project milestone: due Friday, 6 Nov 2026.
- Final report: due Saturday, 5 Dec 2026.

Project Proposal

The project proposal should be one paragraph (200–400 words). Your project proposal should describe:

- What is the problem that you will be investigating? Why is it interesting?
- What reading will you examine to provide context and background?
- What data will you use? If you are collecting new data, how will you do it?
- What method or algorithm are you proposing? If there are existing implementations, will you use them and how? How do you plan to improve or modify such implementations? You don't have to have an exact answer at this point, but you should have a general sense of how you will approach the problem you are working on.
- How will you evaluate your results? Qualitatively, what kind of results do you expect (e.g. plots or figures)? Quantitatively, what kind of analysis will you use to evaluate and/or compare your results (e.g. what performance metrics or statistical tests)?

Submission: Only one student in a group needs to submit the report. Please indicate the names of team members in the report.

Project Milestone

Your project milestone report should be between 2 – 3 pages using the provided template. The following is a suggested structure for your report:

- Title, Author(s)
- Introduction: this section introduces your problem, and the overall plan for approaching your problem
- Problem statement: Describe your problem precisely specifying the dataset to be used, expected results and evaluation
- Technical Approach: Describe the methods you intend to apply to solve the given problem
- Intermediate/Preliminary Results: State and evaluate your results upto the milestone

Submission: Only one student in a group needs to submit the report. Please indicate the names of team members in the report.

Final Report

Your final write-up is required to be between **6 – 8** pages using the provided template, structured like a paper from a computer vision conference. Also, I recommend using LaTex for the final report. Using Overleaf for editing LaTex might be helpful. Please use this template so we can fairly judge all student projects without worrying about altered font sizes, margins, etc. Please indicate if you would like to share share your reports with other students (no impact on grading).

The following is a suggested structure for your report, as well as the rubric that we will follow when evaluating reports. You don't necessarily have to organize your report using these sections in this order, but that would likely be a good starting point for most projects.

- **Title, Author(s)**
- **Abstract**: Briefly describe your problem, approach, and key results. Should be no more than 300 words.
- **Introduction (10%)**: Describe the problem you are working on, why it's important, and an overview of your results
- **Related Work (10%)**: Discuss published work that relates to your project. How is your approach similar or different from others?
- **Data (10%)**: Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, filtering, or other special treatment to use this data in your project?
- **Methods (30%)**: Discuss your approach for solving the problems that you set up in the introduction. Why is your approach the right thing to do? Did you consider alternative approaches? You should demonstrate that you have applied ideas and skills built up

during the quarter to tackling your problem of choice. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.

- **Experiments (30%)**: Discuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate your experimental results.
- **Conclusion (5%)** Summarize your key results – what have you learned? Suggest ideas for future extensions or new applications of your ideas.
- **Writing / Formatting (5%)** Is your paper clearly written and nicely formatted?
- **Supplementary Material**, not counted toward your 6–8 page limit and submitted as a separate file. Your supplementary material might include:
    - Source code (if your project proposed an algorithm, or code that is relevant and important for your project.).
    - Cool videos, interactive visualizations, demos, etc.

  Examples of things to not put in your supplementary material:
    - The entire PyTorch/TensorFlow Github source code.
    - Any code that is larger than 10 MB.
    - Model checkpoints.
    - A computer virus.

Submission: Only one student in a group needs to submit the report. Please indicate the names of team members in the report. You will submit your final report as a PDF and your supplementary material as a separate PDF or ZIP file.

Additional Submission Requirements: We will also ask you do do the following when you submit your project report:

- **Collaborators outside course ARIN 5203 are not allowed.**
- **Any code that was used as a base for projects must be referenced and cited in the body of the paper.** This includes finetuning example code, open-source, or Github implementations. You can use a footnote or full reference/bibliography entry.
- **The project could not be also used in other courses.**